

G. Lejeune Dirichlet's Werke. II.

XVI.

On a Property of Quadratic Forms with Positive Determinant.

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English translation

On a Property of Quadratic Forms with Positive Determinant.

Extract from the proceedings of the Berlin Academy (July 1855), freely translated by the author.

Since the property of quadratic forms which I propose to present in this Note is intimately connected with the theory of the equation

$$t^2 - Du^2 = 1, \tag{1}$$

I shall begin with a few remarks on this equation. In it the given positive integer D is assumed not to be a square, and we shall have to consider only solutions expressed in positive integers t and u .

It is known that, if T, U denote the smallest positive integers satisfying the proposed equation, all positive solutions of the equation are given by the formula

$$(T + U\sqrt{D})^n = t_n + u_n\sqrt{D},$$

where n is to be assigned successively the integer values from $n = 1$ up to $n = \infty$. Among the values thus obtained for u_n , there are infinitely many which are divisible by any given positive integer S , and it is easy to prove that if N denotes the smallest index n for which this condition holds, the other admissible values of n form the sequence of successive multiples of N . To see this, let $D' = DS^2$, and consider the equation

$$t'^2 - D'u'^2 = 1, \tag{2}$$

which has the same form as equation (1). One sees that by putting

$$t = t', \quad u = Su',$$

every solution of equation (2) gives a solution of equation (1) for which u is divisible by S , and conversely. The assertion made above is therefore justified, and one also recognizes that the integer N is given by the equation

$$(T + U\sqrt{D})^N = T' + U'\sqrt{D'},$$

where T' and U' denote the smallest positive values of t' and u' .

The exponent N can be specified without first determining T' and U' . It suffices to know, for each prime divisor p of S , the smallest index ν such that u_ν is divisible by p , and at the same time to know the highest power p^δ of p which divides u_ν . Under the two assumptions just made, one can immediately indicate the exponent of the highest power of p contained in $u_{\nu e}$, where e is an arbitrarily chosen integer. The exponent in question is $\delta + \varepsilon$, where ε , which may also reduce to zero, denotes the highest power of p which divides e .

To recognize the truth of what has just been stated, let θ and η be two integers satisfying equation (1), and let p^k , with $k \neq 0$, be the highest power of p contained in η . Deriving a new solution from this one by means of the formula

$$(\theta + \eta\sqrt{D})^m = t + u\sqrt{D},$$

we shall have

$$u = \eta \left(m\theta^{m-1} + \frac{m(m-1)(m-2)}{1 \cdot 2 \cdot 3} \theta^{m-3} \eta^2 D + \dots \right).$$

It is manifest, since θ is not divisible by p , that the highest power of p contained in u will be p^k if m is not divisible by p , and will be p^{k+1} if $m = p$. Since raising to an arbitrary power can be reduced to a combination of the two preceding special cases, the result stated above is established.

From the property just proved, we see that the necessary and sufficient condition for $u_{\nu e}$ to be divisible by an arbitrary power p^α , where $\alpha > 0$, is that e have the factor $p^{\alpha-\delta}$, the exponent $\alpha - \delta$, if it becomes negative, being replaced by zero.

Let us now use indices to denote the unequal prime numbers p contained in S , and the corresponding values of ν and δ , and put

$$S = p_1^{\alpha_1} p_2^{\alpha_2} \dots$$

The required exponent N , according to what precedes, will be the least common multiple of the numbers

$$\nu_1 p_1^{\alpha_1 - \delta_1}, \quad \nu_2 p_2^{\alpha_2 - \delta_2}, \quad \dots$$

It is easy to see that, if without introducing new prime factors into S one lets all the exponents $\alpha_1, \alpha_2, \dots$ increase indefinitely, the quotient S/N will soon cease to vary, while retaining a value independent of $\alpha_1, \alpha_2, \dots$

The theorem just established for equation (1) leads to a remarkable consequence concerning the theory of quadratic forms with positive determinant. To the preceding assumptions, add that D should have no square divisor. Let h denote the number of classes into which the quadratic forms with common determinant D are distributed, and give h' the analogous meaning relative to the determinant $D' = DS^2$. We then have the equation

$$h' = h \frac{\log(T + U\sqrt{D})}{\log(T' + U'\sqrt{D'})} SR,$$

established in my memoir *Researches on Various Applications of Infinitesimal Analysis to the Theory of Numbers*, part two. As for the quantity R which enters into it and which we have not had to mention above, it will suffice for our purpose to recall that this quantity does in fact depend on the primes $p_1, p_2, \dots$, but in no way on the exponents $\alpha_1, \alpha_2, \dots$. Since our equation can be put in the form

$$h' = h \frac{S}{N} R,$$

we conclude from the preceding result that from every positive determinant D one can derive infinitely many other determinants of the form DS^2 , all having the same number of classes. By choosing the integer D and the primes $p_1, p_2, \dots$ suitably, one can arrange, in infinitely many ways, that the invariable number of classes coincide for this entire infinite series of determinants with the number of genera. This confirms the conjecture stated by the illustrious author of the *Disquisitiones arithmeticae* (see art. 304), according to which the positive determinants possessing a single class in each of the corresponding genera form an infinite sequence. This is an analytical phenomenon all the more remarkable because the negative determinants enjoying the same property appear to be only finite in number, and indeed very limited. If one may rely

on an induction carried very far – up to 10000, if I am not mistaken – first by Euler, who dealt with these same numbers under the name *numeri idonei*, that is, numbers suitable for finding very large prime numbers, at a time when the theory of quadratic forms, founded by Lagrange, did not yet exist, and later by Gauss, the negative determinants in question would number only 65, and the greatest of these determinants, disregarding the sign, would have the value 1848.